SAMPLE COMMERCIAL BUILDING

Electrical Design Calculation Report

Cable sizing · Voltage drop · Short circuit · Earthing

Project	Sample Commercial Building
Project no.	P-2026-014
Client	Sample Developments LLC
Consultant	Acme Engineering Consultants
Contractor	*Not defined*
Document no.	ELEC-CAL-001
Revision	P01 — 15 Oct 2026
Issue status	For review

	Name	Signature	Date
Prepared by	A. Engineer		
Checked by	*Not defined*		
Approved by	*Not defined*		

MOCKUP — illustrative values only. Calculated checks against the project limits — engineering review and approval are separate. Fields marked *Not defined* must be completed before issue.

DOCUMENT CONTROL AND CHECK SUMMARY

Revision history

Rev.	Date	Description	By
P01	15 Oct 2026	First issue for review	AE

Calculation record

Results calculated	15 Oct 2026 10:42 — **current** (inputs unchanged since)
Standards	IEC 60364-5-52:2009 · IEC 60909-0:2016 · IEC 60364-5-54:2011
Cable data	IEC 60364-5-52 Table B.52 (reference data — not manufacturer-verified)
Application	LV Design Studio v1.39 · snapshot 7F3A-21C0

Check summary

142	3	1	4	2
PASS	WARNING	FAIL	NOT CHECKED	DATA REQUIRED

Items needing attention (all listed — none omitted)

Status	Item	Finding	Ref.
FAIL	F-02	Total voltage drop 5.20 % exceeds the 4.0 % project limit	3.2
WARNING	SMDB-1	Busbar loading 92 % of rating (limit 90 %)	4.1
WARNING	F-07	Disconnection time 0.38 s, margin under 10 %	5.3
WARNING	DB-G2	Spare ways below 20 %	2.4
DATA REQUIRED	TX-1	Transformer impedance not entered	6.1
DATA REQUIRED	ATS-1	Manufacturer withstand rating not entered	6.2

3 CABLE SIZING AND VOLTAGE DROP

3.2 Feeder voltage drop

Voltage drop per IEC 60364-5-52 Annex G, using the cable's mV/A/m at the operating temperature. Totals include the upstream drop from the transformer LV terminals.

Table 7 — Feeder voltage drop

Circuit	From → To	Cable	Method	Ib A	In A	Iz A	Length m	VD %	Total VD %	Limit %	Status
F-01	MDB → SMDB-1	4C × 35 mm² Cu XLPE	E	72.0	80	126.0	45	0.95	2.10	4.0	PASS
F-02	MDB → DB-R1	4C × 16 mm² Cu XLPE	C	63.0	63	81.0	80	4.05	5.20	4.0	**FAIL**
F-03	SMDB-1 → DB-G1	4C × 10 mm² Cu XLPE	B2	31.5	40	52.0	22	0.90	3.00	4.0	PASS
F-04	SMDB-1 → DB-G2	4C × 6 mm² Cu PVC	B2	28.0	32	36.0	30	1.84	3.94	4.0	**WARNING**
F-05	MDB → PFC-1	4C × 95 mm² Cu XLPE	E	—	250	298.0	8	—	—	4.0	*NOT CHECKED*
Worst case									**5.20**	**4.0**	**FAIL**

FAIL F-02: the total voltage drop is 5.20 %, above the 4.0 % project limit. Increase the cable to 4C × 25 mm² or reduce the route length.

REFERENCE DATA Cable ratings are from IEC 60364-5-52 Table B.52. Confirm them against the selected manufacturer's data before construction.

Status key: PASS — within limit · **WARNING** — within limit, small margin · **FAIL** (boxed) — outside limit · *NOT CHECKED* — nothing to check against · *DATA REQUIRED* — input missing.

7 CABLE SCHEDULE

Table 12 — Cable schedule (header repeats on every page)

Cable ref.	From	To	Cores × size mm²	Type	CPC mm²	Method	Length m	Ib A	Iz A	Ca·Cg	Ik max kA	Zs Ω	Status
C-001	TX-1	MDB	4 × (1C × 240)	Cu XLPE/SWA	120	F	12	1 250.0	1 520.0	0.87	31.40	0.008	PASS
C-002	MDB	SMDB-1	4C × 35	Cu XLPE/SWA	16	E	45	72.0	126.0	0.91	18.20	0.061	PASS
C-003	MDB	DB-R1	4C × 16	Cu XLPE/SWA	16	C	80	63.0	81.0	0.94	18.20	0.214	**FAIL**
C-004	SMDB-1	DB-G1	4C × 10	Cu XLPE	10	B2	22	31.5	52.0	0.80	9.60	0.188	PASS
C-005	SMDB-1	DB-G2	4C × 6	Cu PVC	6	B2	30	28.0	36.0	0.80	9.60	0.352	**WARNING**
C-006	MDB	PFC-1	4C × 95	Cu XLPE/SWA	50	E	8	—	298.0	1.00	31.40	—	*NOT CHECKED*
C-007	MDB	ATS-1	4C × 120	Cu XLPE/SWA	70	E	15	210.0	344.0	1.00	31.40	0.021	*DATA REQUIRED*

Units in the column heads; figures rounded to 2 decimals (lengths and protective-device ratings whole numbers). Narrow tables print at 8 pt and are never shrunk below 7 pt — wide tables move to A3 or split into a continuation table.